
MI 101: Thermodynamics

Course Code	MI 101
Course Name	Thermodynamics
L-T-P-C	2-1-0-3
Prerequisites	NIL Sem. Both
Approval	5th Senate (Oct 2013)

Course Contents

- **Introduction:** Introduction to thermodynamic system, surrounding, state, process, properties, equilibrium, heat and work, Zeroth Law of Thermodynamics;
- **Properties of Pure Simple Compressible Substance:** PvT Surface, Pv, Tv, TP diagrams. Equation of state for ideal and real gases. Virial equation of state, Van-der waal, Redlich- Kwong, Peng Robinson equation of state etc. use of steam tables and Mollier diagram;
- **First Law of Thermodynamics:** First law application to non-flow processes such as isochoric, isobaric, isothermal, adiabatic and polytropic processes. Steady flow energy equation, flow work. Application to various practical systems viz nozzles, diffuser, turbines, heat exchangers etc. Application of energy equation to transient flow problems;
- **Second Law of Thermodynamics:** Second law, reversible and irreversible processes, Clausius and Kelvin Planck statements, Carnot cycle, corollaries of second law: thermodynamic temperature scale, Clausius inequality, entropy as a property, principle of increase of entropy. Calculation of entropy change;
- **Thermodynamic Cycles:** Otto, Diesel, Rankine cycles and their applications. Vapour compression refrigeration cycle.

References

1. Van Wylen and Sonntag, **Fundamentals of Thermodynamics**, John Wiley.
2. Roger G. F. C. and Mayhew, **Engineering Thermodynamics Work and Heat Transfer**, Longman.
3. Smith J. M., Van Ness H. C. and Abbott M. M, **Introduction to Chemical Engineering Thermodynamics**, Tata McGraw Hill.
4. Version S. I., Moran M. J. and Shapiro H. M., **Fundamentals of Engineering Thermodynamics**, John Wiley.